

Small intestine

Department of Anatomy

The abdomen (5)

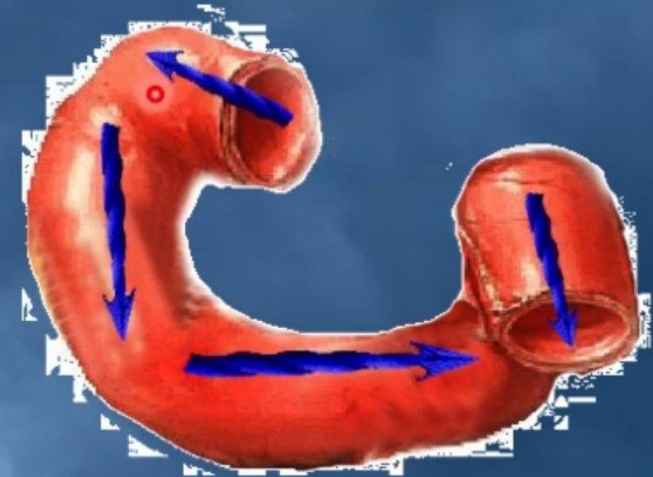
Second Part

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Lecture Points

- **Structure of intestine**
- **Five main functions of the bowel (intestine).**
- **Parts of the Small Intestine.**
- **The Duodenum (Location and Description)**
 - ✓ **Parts.**
 - ✓ **Length.**
 - ✓ **Direction.**
 - ✓ **Relation to Peritoneum.**
 - ✓ **First part of Duodenum.**
 - ✓ **Second part of Duodenum**
 - ✓ **Third part of Duodenum**
 - ✓ **Fourth part of Duodenum.**
 - ✓ **Nerve Supply of Duodenum**
 - ✓ **Arterial Supply of Duodenum**
 - ✓ **Veins drainage of Duodenum.**
 - ✓ **Lymph drainage of duodenum.**
- **The Jejunum & Ileum.**
- **Jejunum and Ileum (Location and Description).**
- **The root of mesentery of the small intestine**
- **The duodenojejunal flexure.**
- **The ileocecal junction.**
- **The anatomical features of Jejunum & Ileum**
- **Difference Between Jejunum & Ileum.**
- **Blood Supply of Jejunum & Ileum.**
- **Nerve Supply of Jejunum & Ileum**
- **Lymph drainage of Jejunum & Ileum**

The duodenum is divided into four parts



1) Superior (First) Part

- It has a mobile or free section, termed the **duodenal cap** (because of its appearance on radiographs), into which **the pylorus invaginates**.

2) Descending (Second) Part

- It contains the junction of **the foregut** and **midgut**, where the common bile and main pancreatic ducts open.
- It contains the **greater papilla**, on which **terminal openings of the bile** and **main pancreatic ducts** are located, and **the lesser papilla**, which lies **2 cm above the greater** papilla and marks the site of entry of the **accessory pancreatic duct**.

The duodenum is divided into four parts



3. Transverse (Third) Part

- It is **the longest part** and **crosses the IVC, aorta, and vertebral column** to the left.
- It is crossed **anteriorly** by the **superior mesenteric vessels**.

4. Ascending (Fourth) Part

- Ascends to the left of **the aorta** to the level of **the second lumbar vertebra** and **terminates at the duodenojejunal junction**, which is fixed in position by **the suspensory ligament (of Treitz)**, a surgical landmark.

First part of Duodenum

Relations

The second inch of the duodenum:

- **Anteriorly:**

- a. The quadrate lobe of the liver,
- b. The neck of the gall bladder

- **Posteriorly:**

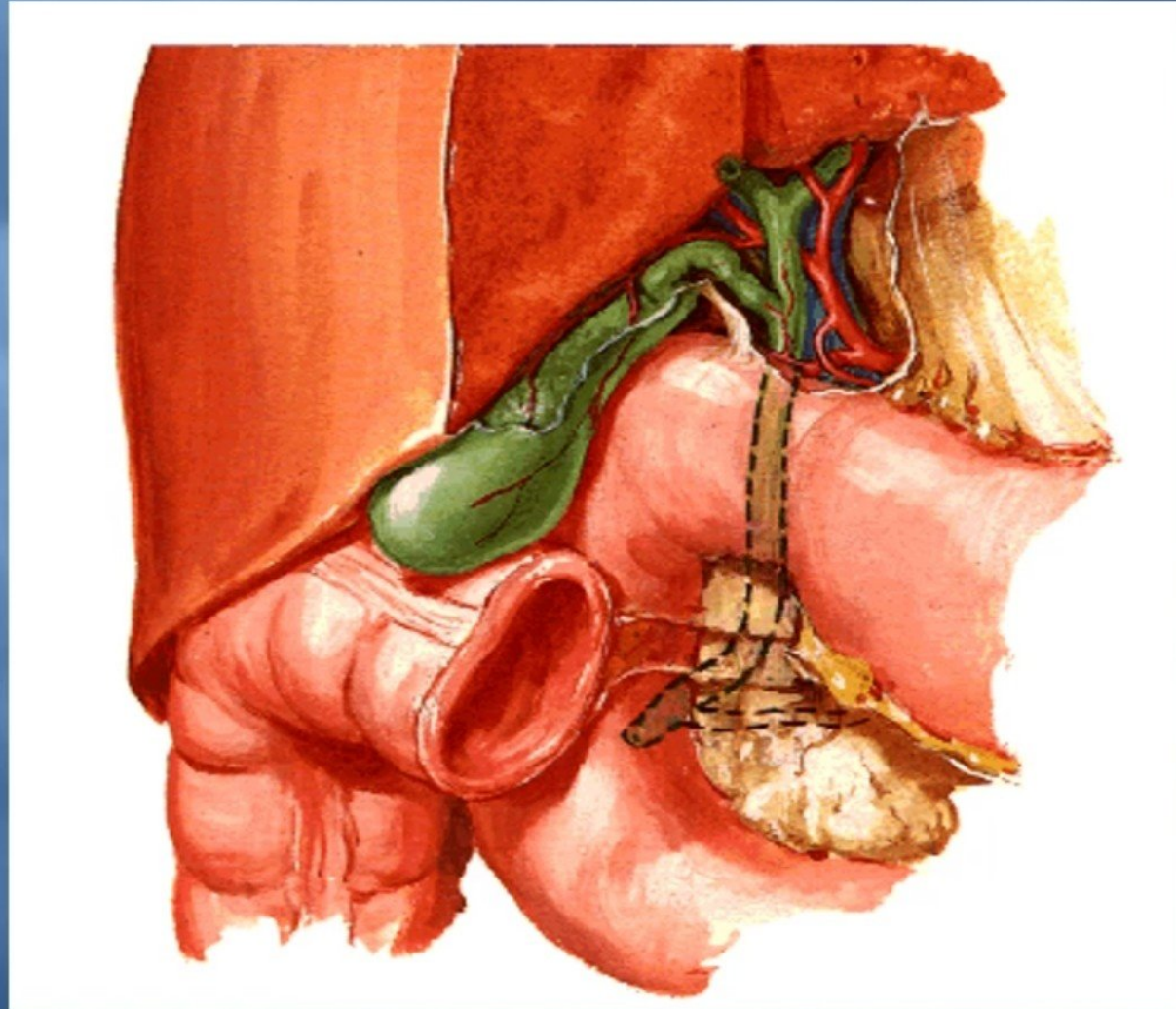
- a. The bile duct.
- b. The gastroduodenal artery.
- c. The portal vein.
- d. The inferior vena cava.

- **Superiorly:**

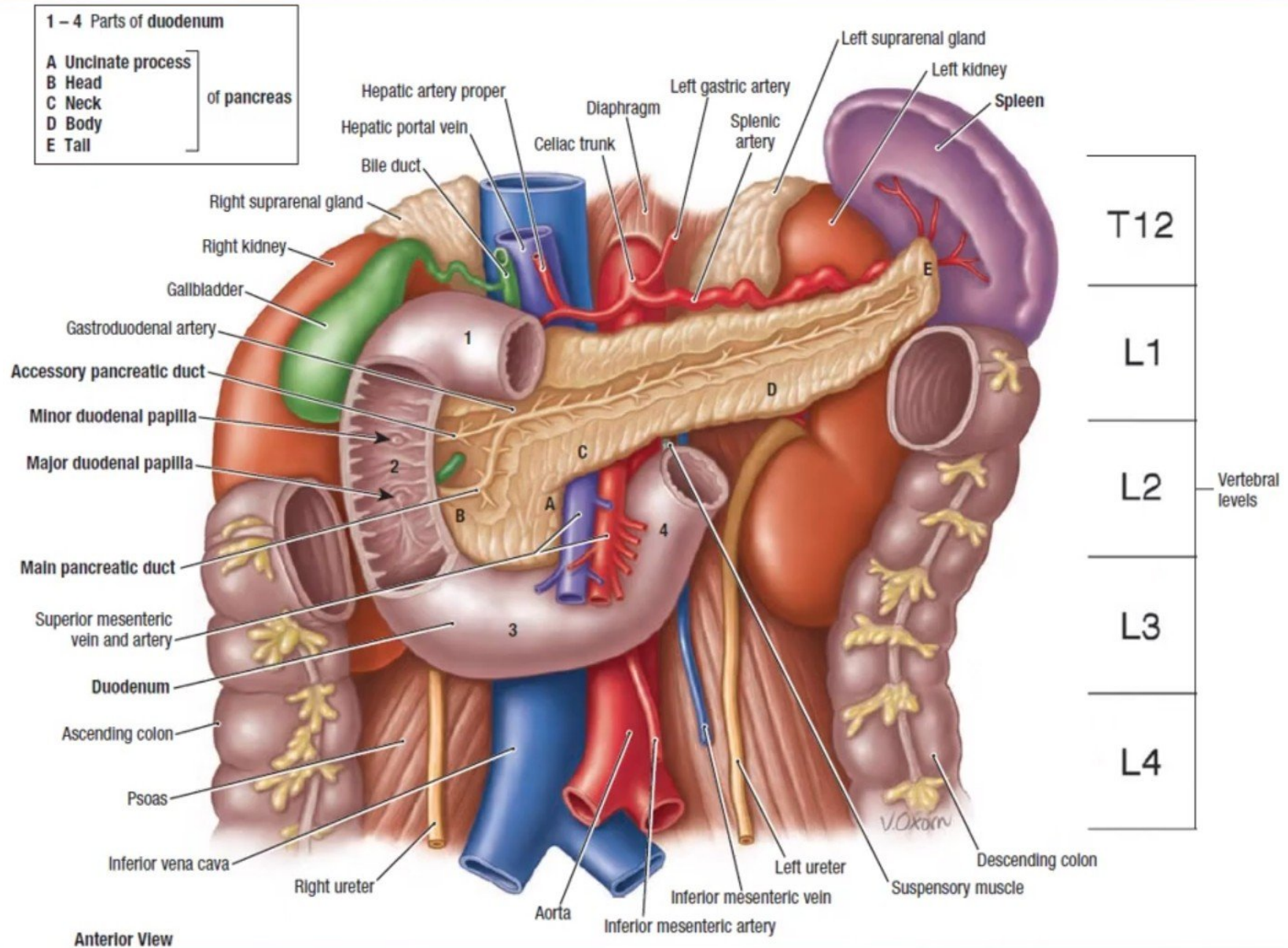
the opening into the lesser sac.

- **Inferiorly:**

The pancreas.



First part of Duodenum



Second part of Duodenum

- **Length:** three inches long.
- **Extent:** from the neck of gall bladder to L3
- **Direction:** descends vertically.
- **Peritoneal covering:** covered by the peritoneum only on the anterior surface except the middle part where is crossed by transverse colon.



Second part of Duodenum

Relations of the second part:

• Anteriorly:

1. Upper part: right lobe of the liver.
2. The middle part: transverse colon.
3. Lower part: loops of the jejunum.

• Laterally:

The right colic flexure.

The fat in front of the right kidney.

• Medially:

The head of pancreas

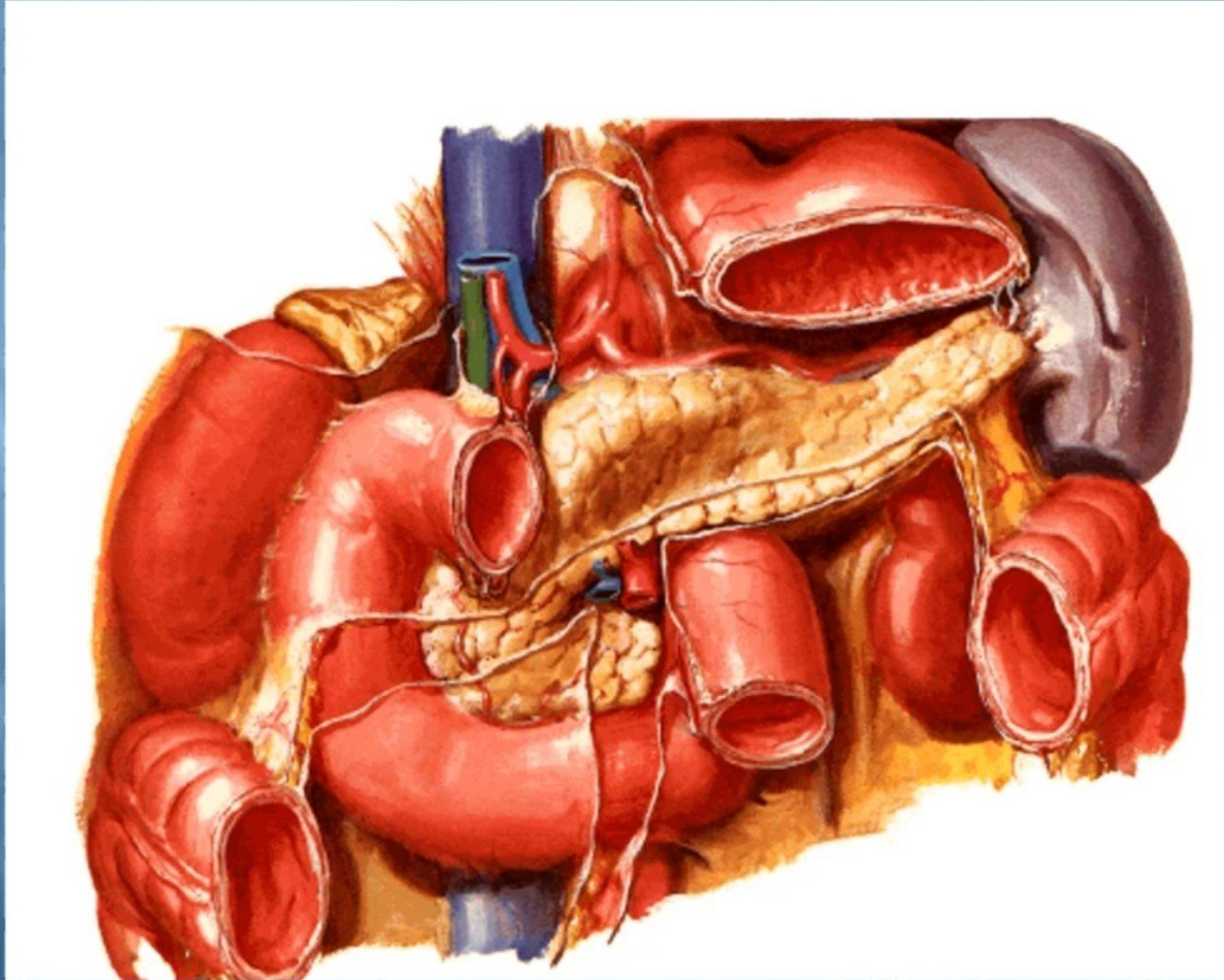
The bile duct

• Posteriorly:

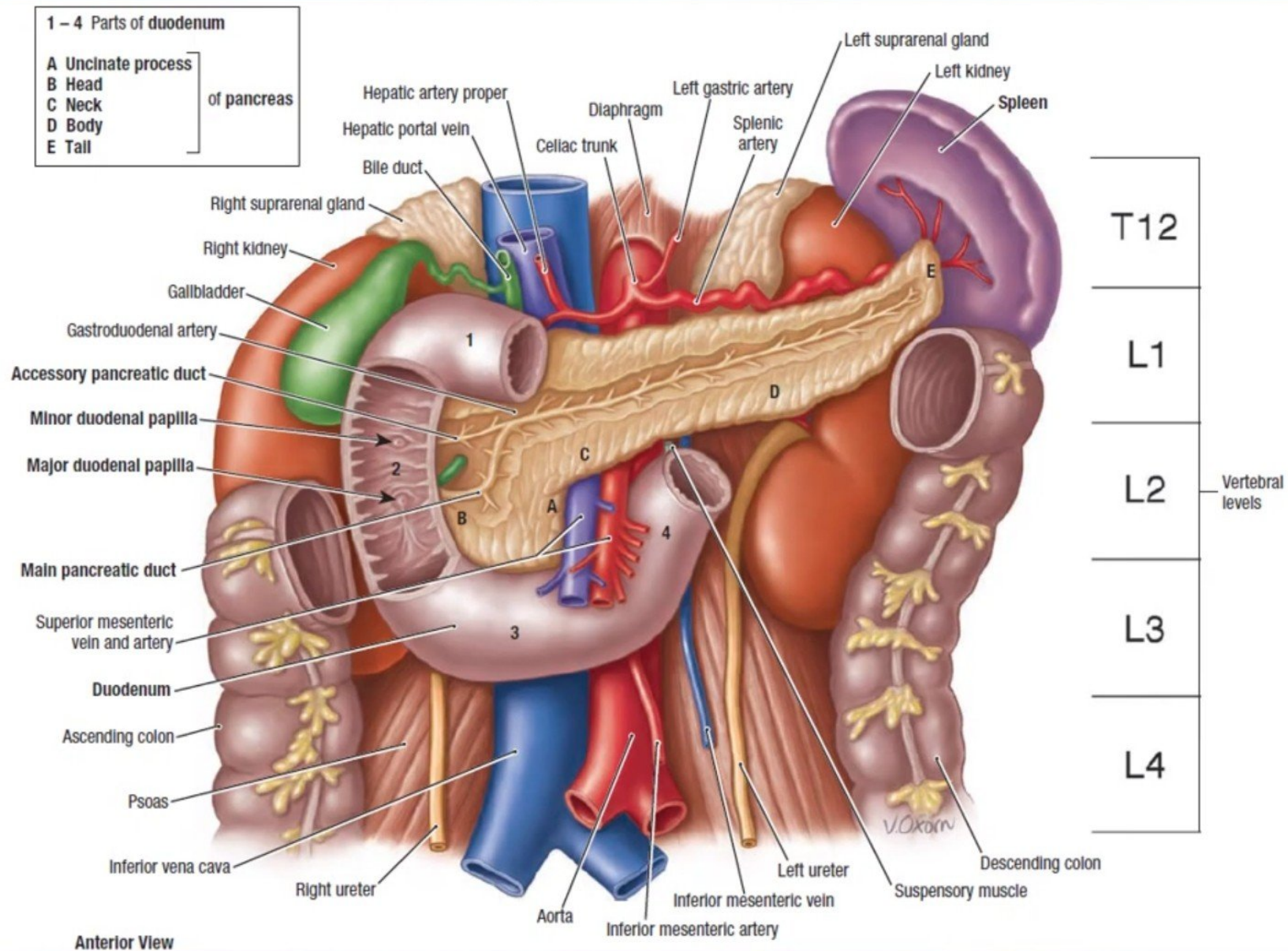
The hilum of right kidney.

The right renal vessels.

The right psoas major muscle.

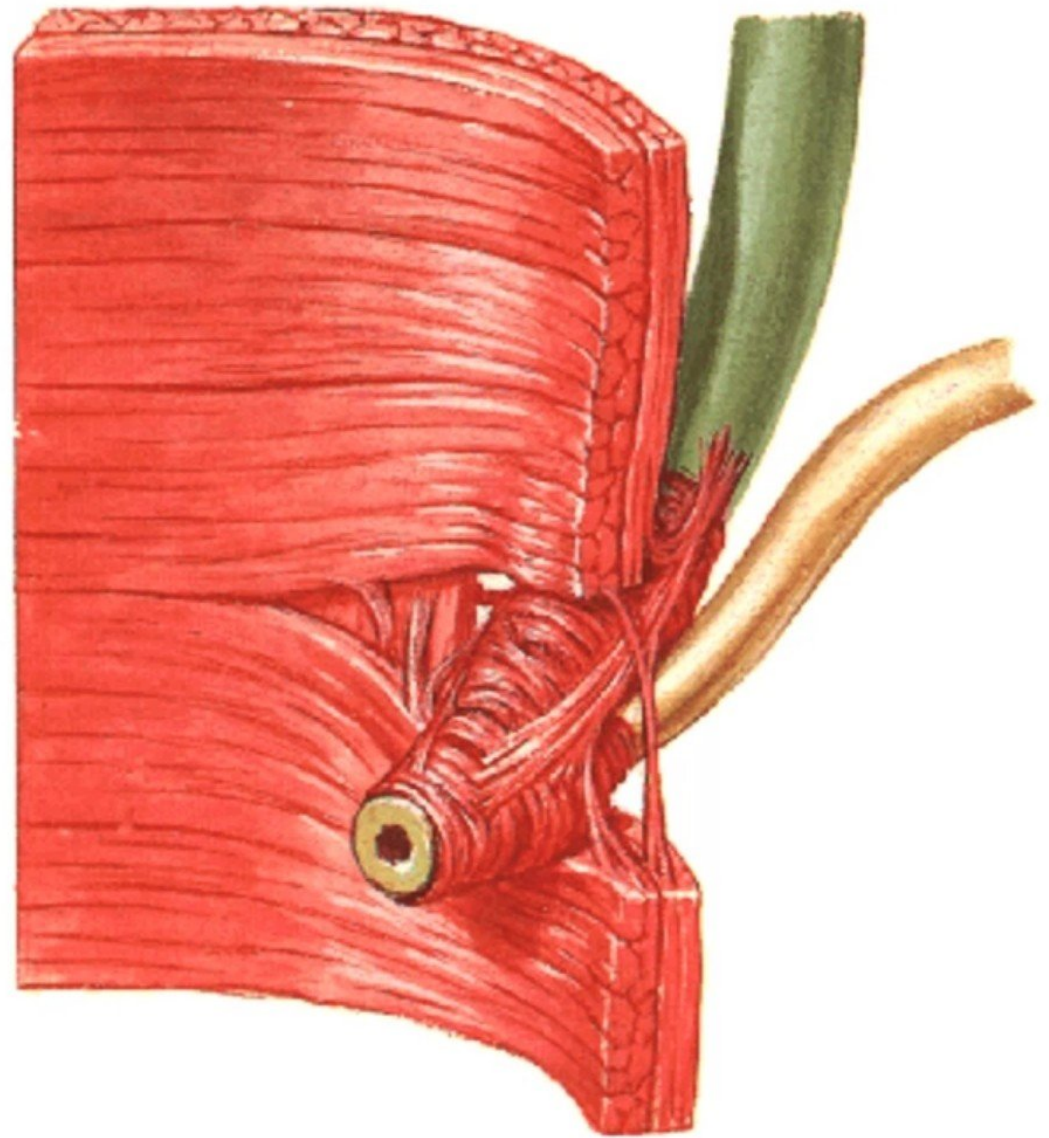


Second part of Duodenum

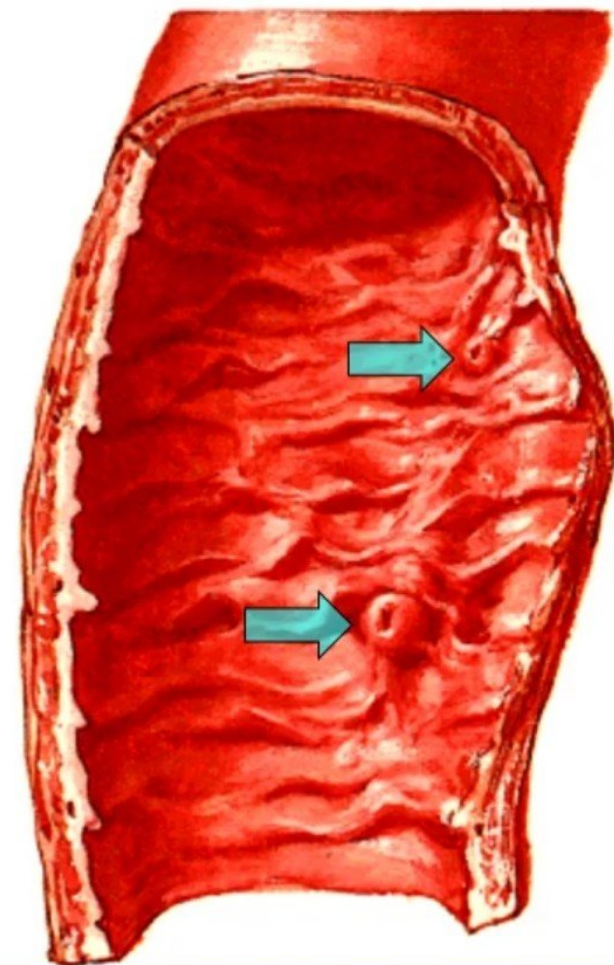
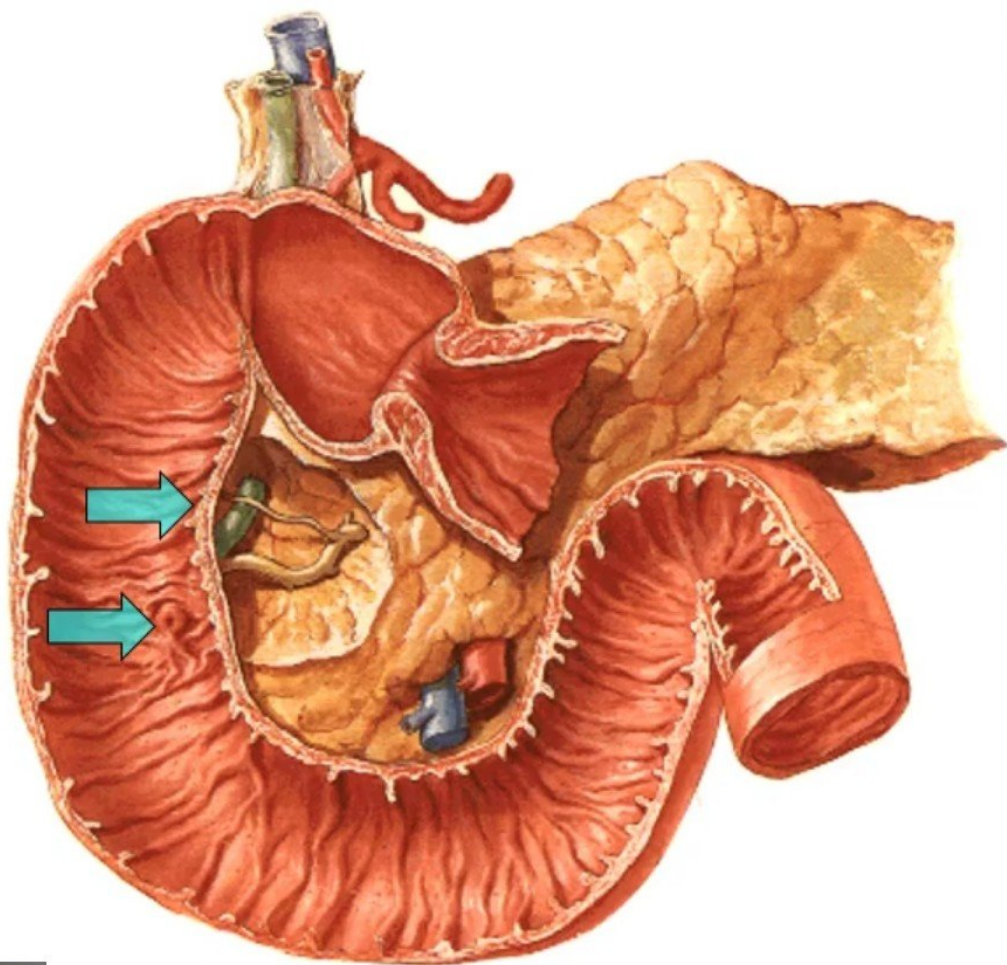


Structures Opening in the 2nd Part of the Duodenum

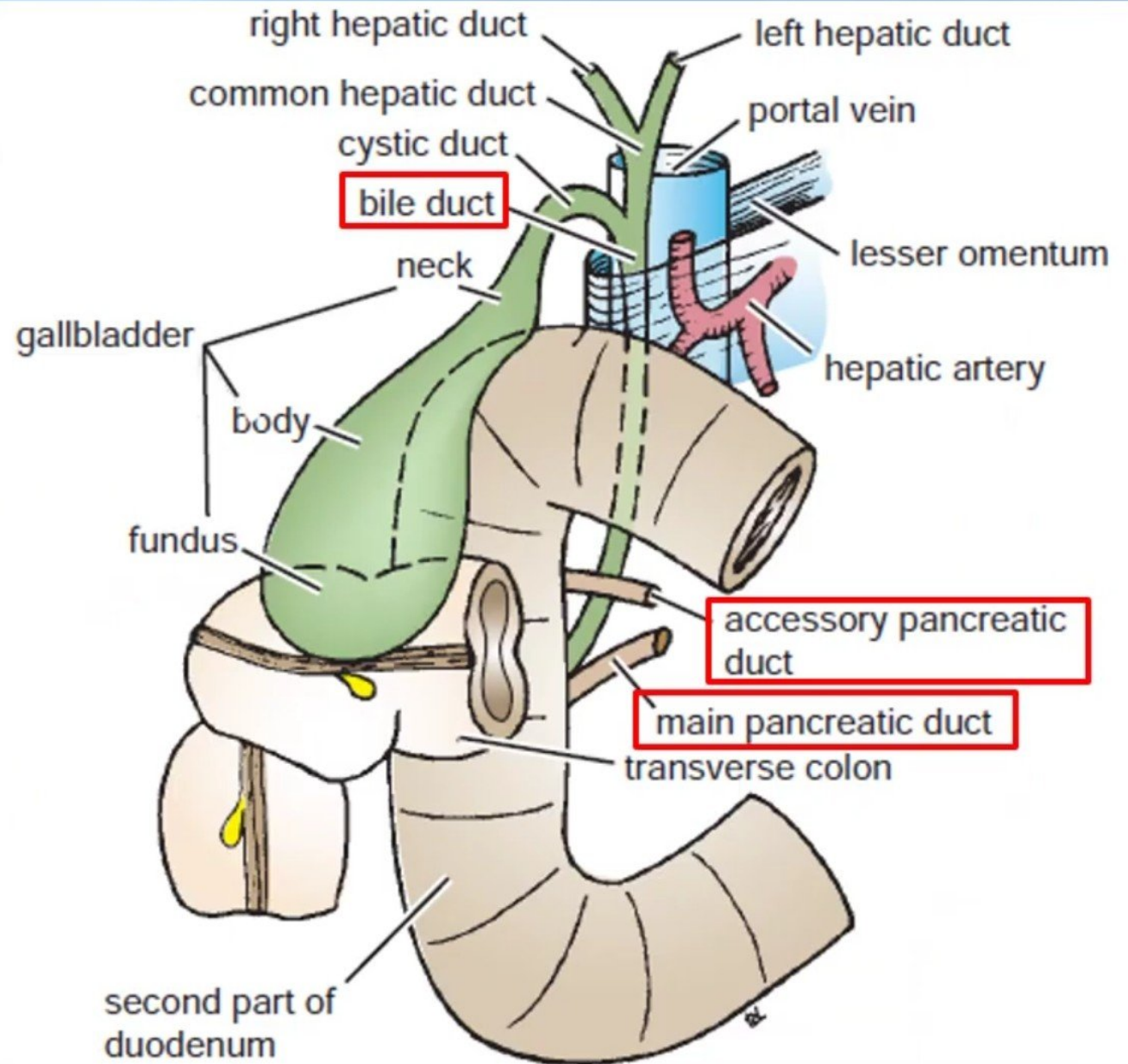
- ❖ The **bile duct unites with the pancreatic duct** forming a dilatation called the **hepatopancreatic ampulla** (ampulla of Vater).
- ❖ The ampulla opens on an elevation called the **major duodenal papilla**.
- ❖ The **accessory pancreatic duct opens** one-inch **above the major duodenal papilla**, forming a smaller elevation called the **minor duodenal papilla**.



Structures Opening in the 2nd Part of the Duodenum

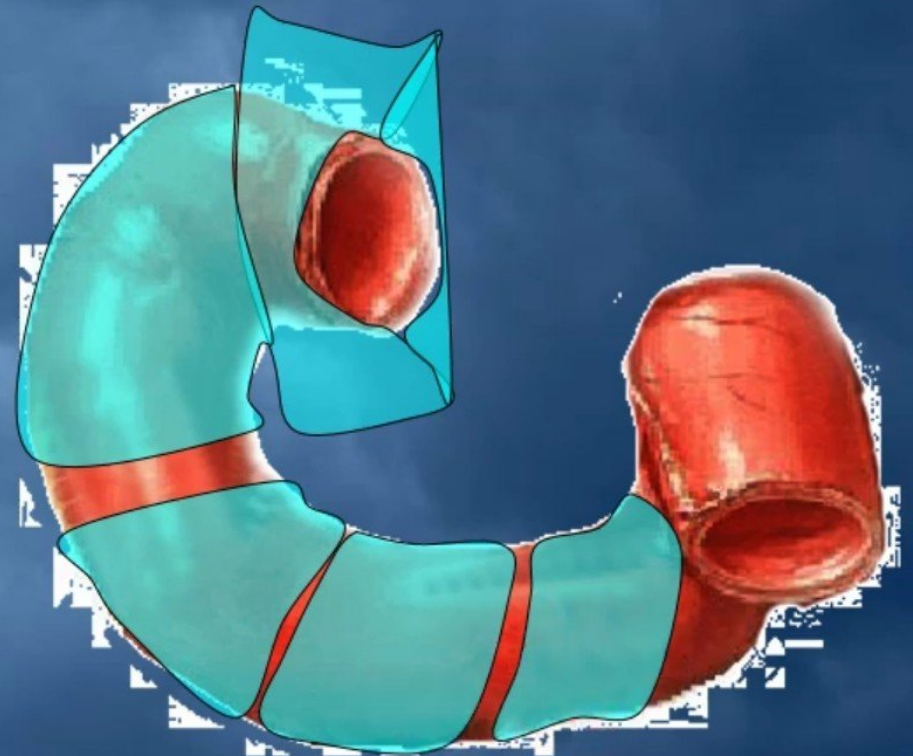


Structures Opening in the 2nd Part of the Duodenum



Third part of Duodenum

- **Length:** four inches.
- **Extent:** at the level of the 3rd lumbar vertebra.
- **Direction:** lies in a horizontal plane.
- **Peritoneal covering:** is only covered by peritoneum anteriorly and inferiorly except the site of attachment of mesentry.



Third part of Duodenum

Relations of the 3rd part:

Anteriorly:

1. The root of the mesentery.
2. The coils of jejunum.

Posteriorly:

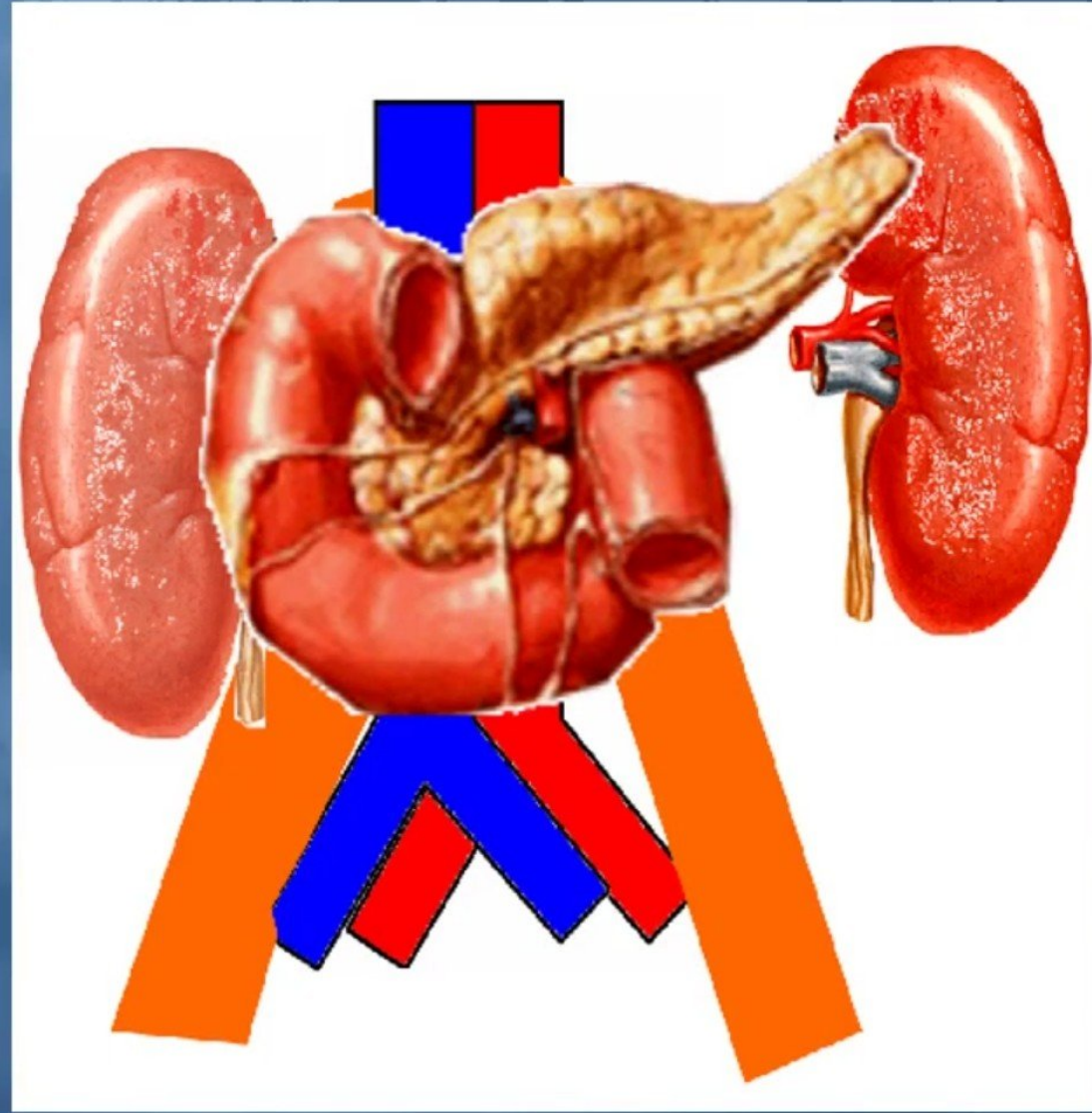
1. The right ureter.
2. The right psoas major muscle.
3. The right testicular (or ovarian) vessels.
4. Inferior vena cava.
5. Abdominal aorta & origin of the inferior mesenteric artery.

Superiorly:

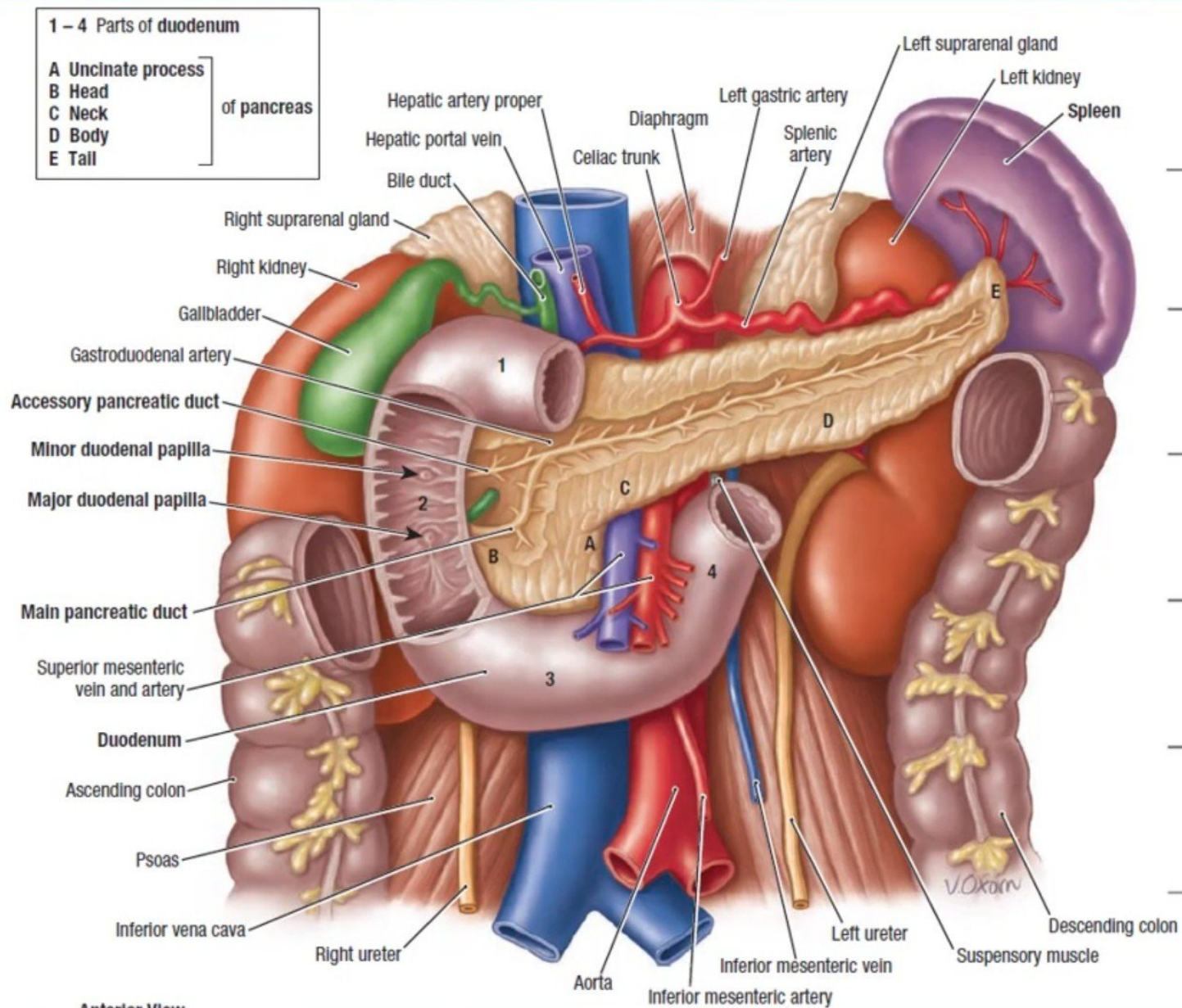
The pancreas.

Inferiorly:

Coils of the jejunum.

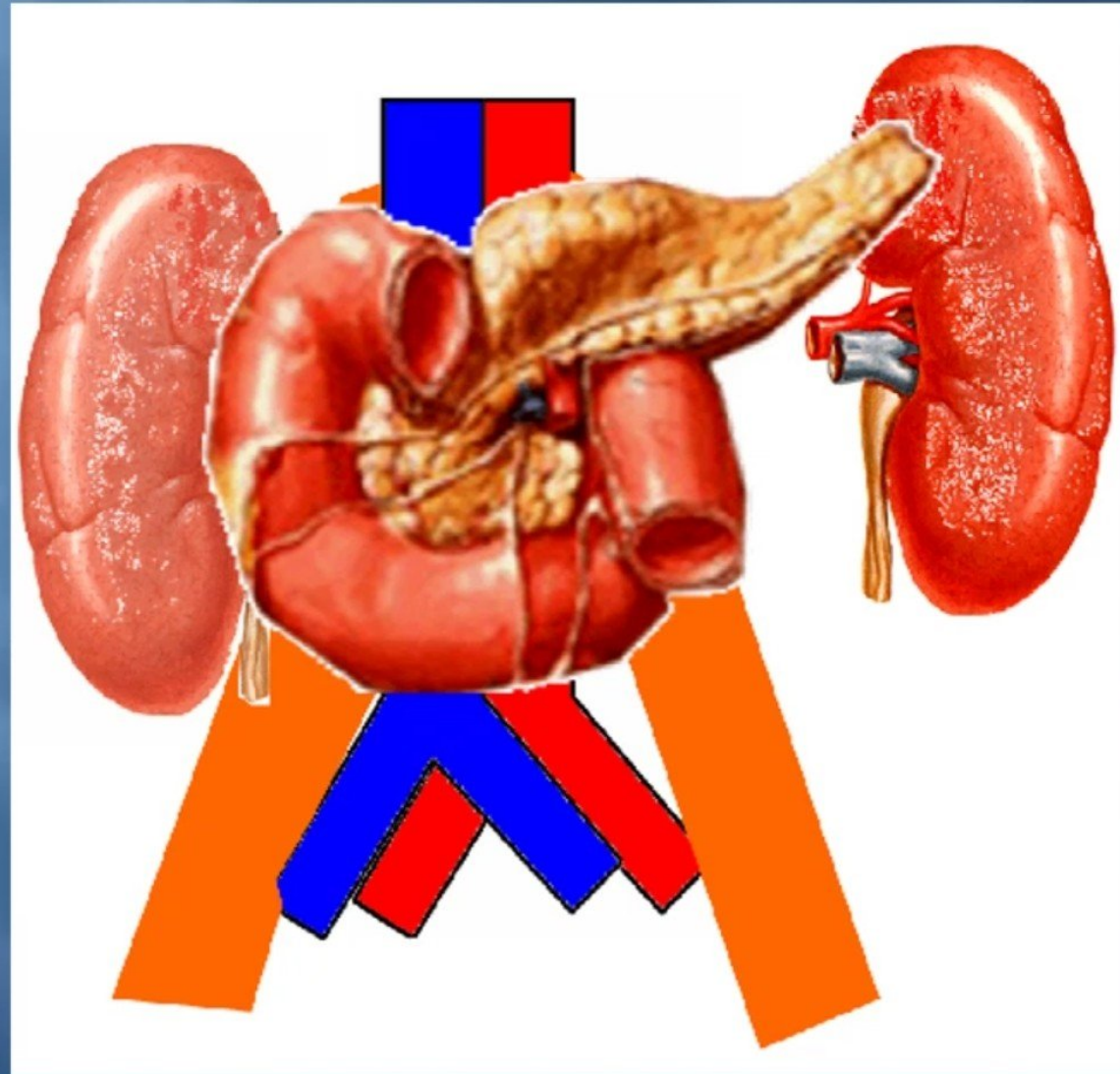


Third part of Duodenum

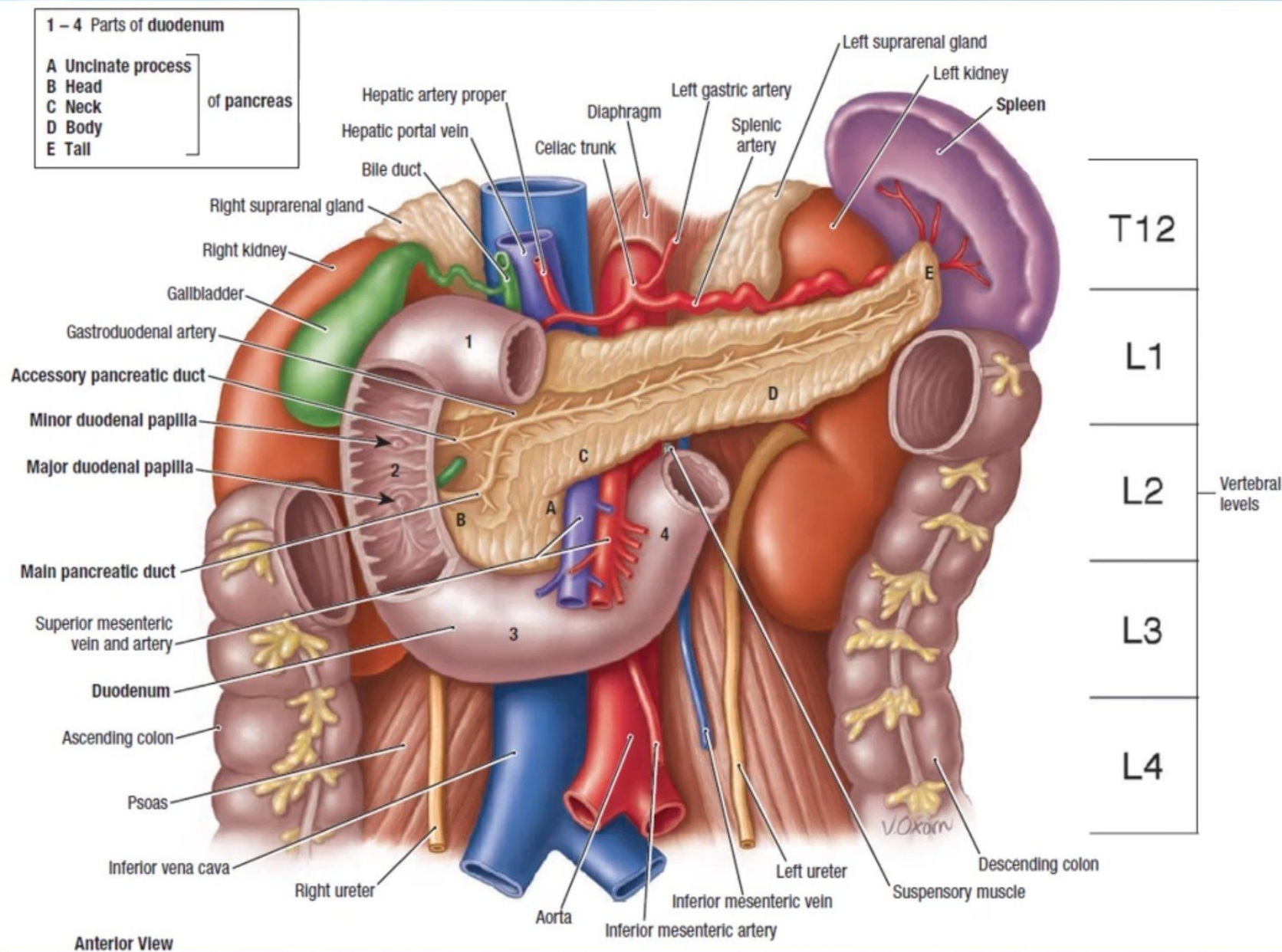


Fourth part of Duodenum

- **Length:** one inch long.
- **Extent:** from the level of the 3rd to the level of the 2nd lumbar vertebrae.
- **Direction:** ascends to end by forming the duodenojejunal flexure.
- **Peritoneal covering:** is covered by the peritoneum anteriorly and to the left.



Fourth part of Duodenum



Summary of Duodenum relations

Part of Duodenum	Anterior	Posterior	Medial	Superior	Inferior	Vertebral Level
Superior (1st part)	Peritoneum Gallbladder Quadrant lobe of liver	Bile duct Gastroduodenal artery Hepatic portal vein IVC		Neck of gallbladder	Neck of pancreas	Anterolateral to L1 vertebra
Descending (2nd part)	Transverse colon Transverse mesocolon Coils of small intestine	Hilum of right kidney Renal vessels Ureter Psoas major	Head of pancreas Pancreatic duct Bile duct			Right of L2–L3 vertebrae
Inferior (horizontal or 3rd part)	Superior mesenteric artery Superior mesenteric vein Coils of small intestine	Right psoas major IVC Aorta Right ureter		Head and uncinate process of pancreas Superior mesenteric artery and vein		Anterior to L3 vertebra
Ascending (4th part)	Beginning of root of mesentery Coils of jejunum	Left psoas major Left margin of aorta	Superior mesenteric artery and vein	Body of pancreas		Left of L3 vertebra

Nerve Supply of Duodenum

- The nerves are derived from **sympathetic** and **parasympathetic (vagus) nerves** from **the celiac** and **superior mesenteric plexuses**.

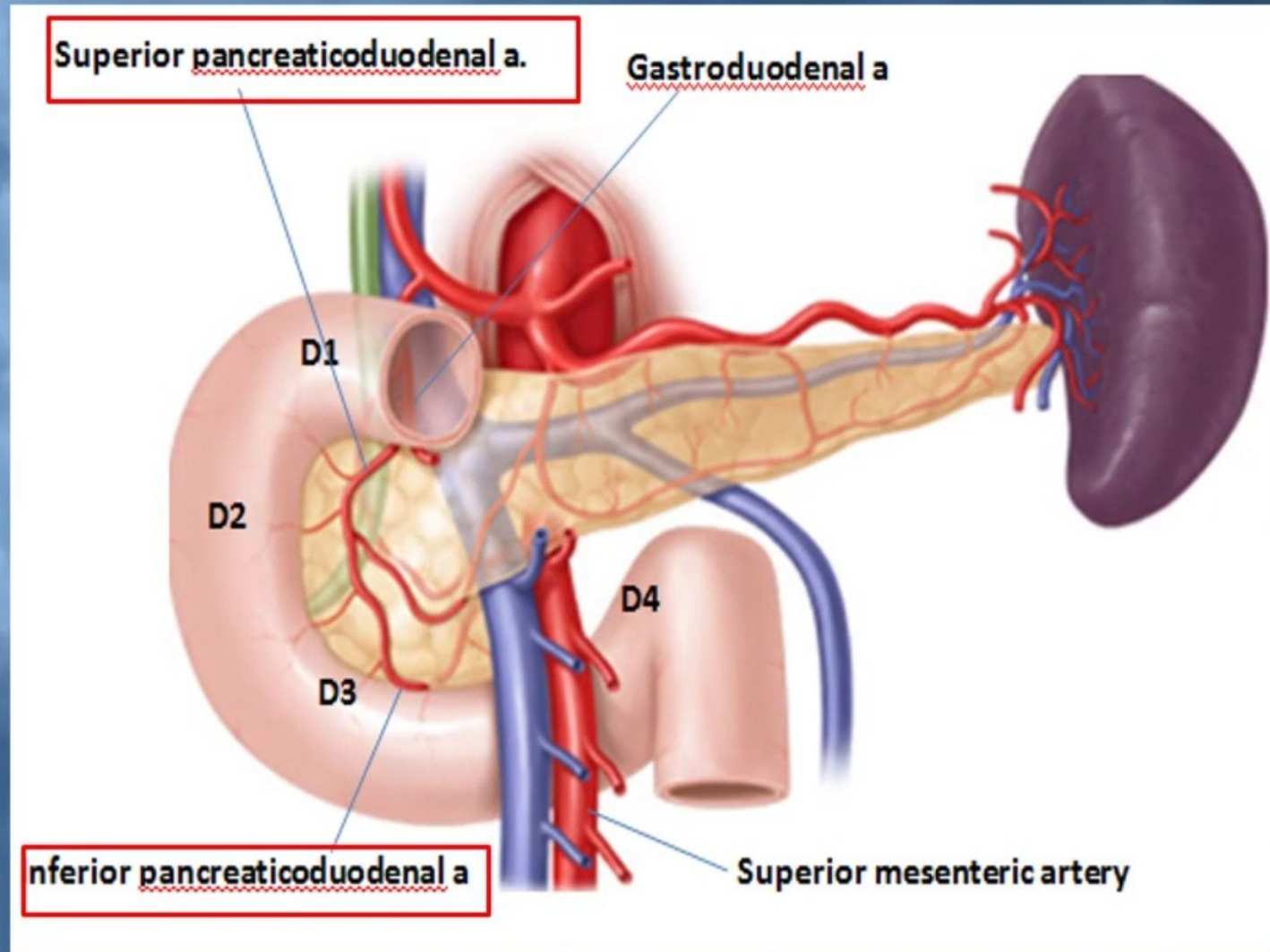
Arterial Supply of Duodenum

1- Upper half :

(1st part + upper 1/2 of 2nd part) is supplied by **the superior pancreaticoduodenal artery**, a branch of the **gastrooduodenal artery**.

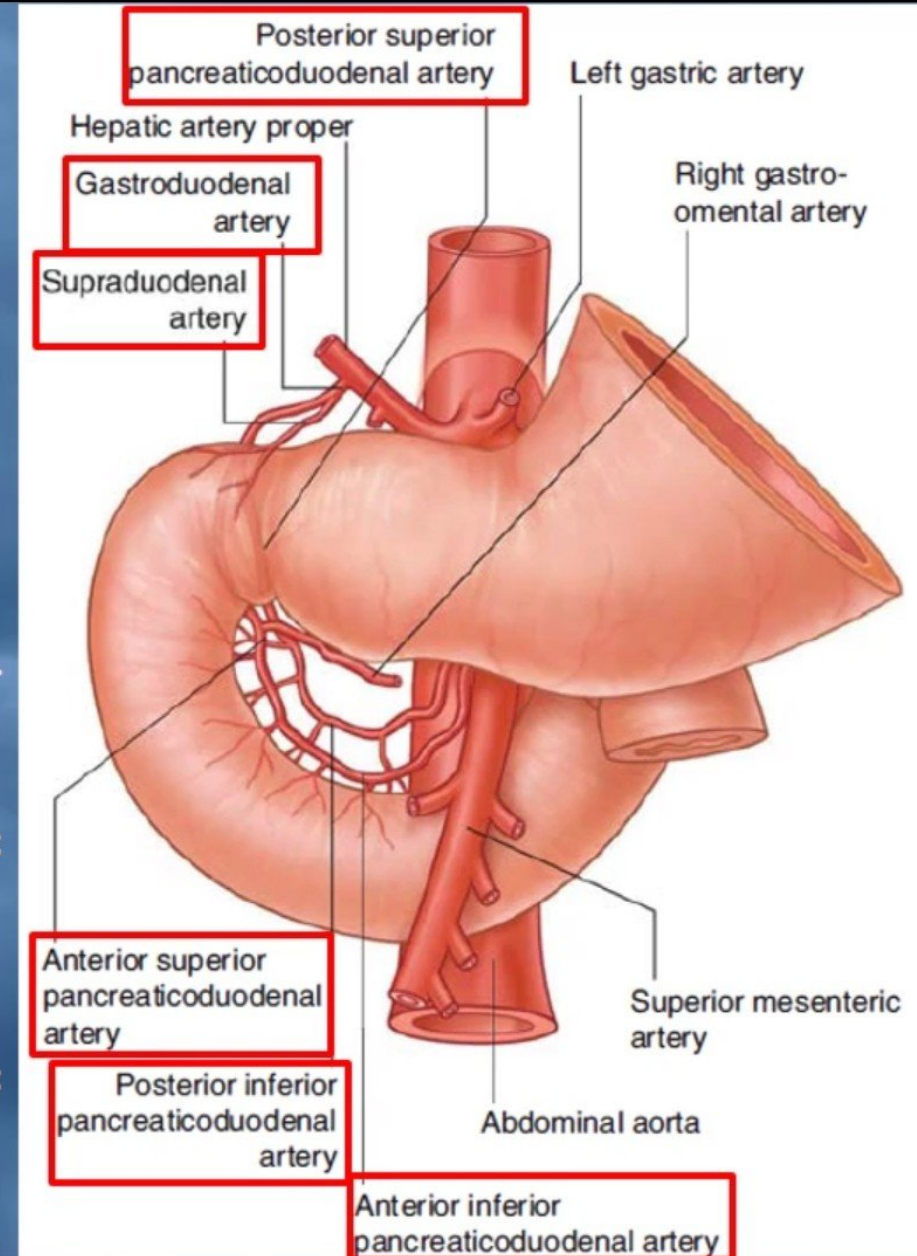
2- Lower half :

(lower 1/2 of 2nd part + 3rd + 4th part) is supplied by **the inferior pancreaticoduodenal artery**, a branch of the **superior mesenteric artery**.



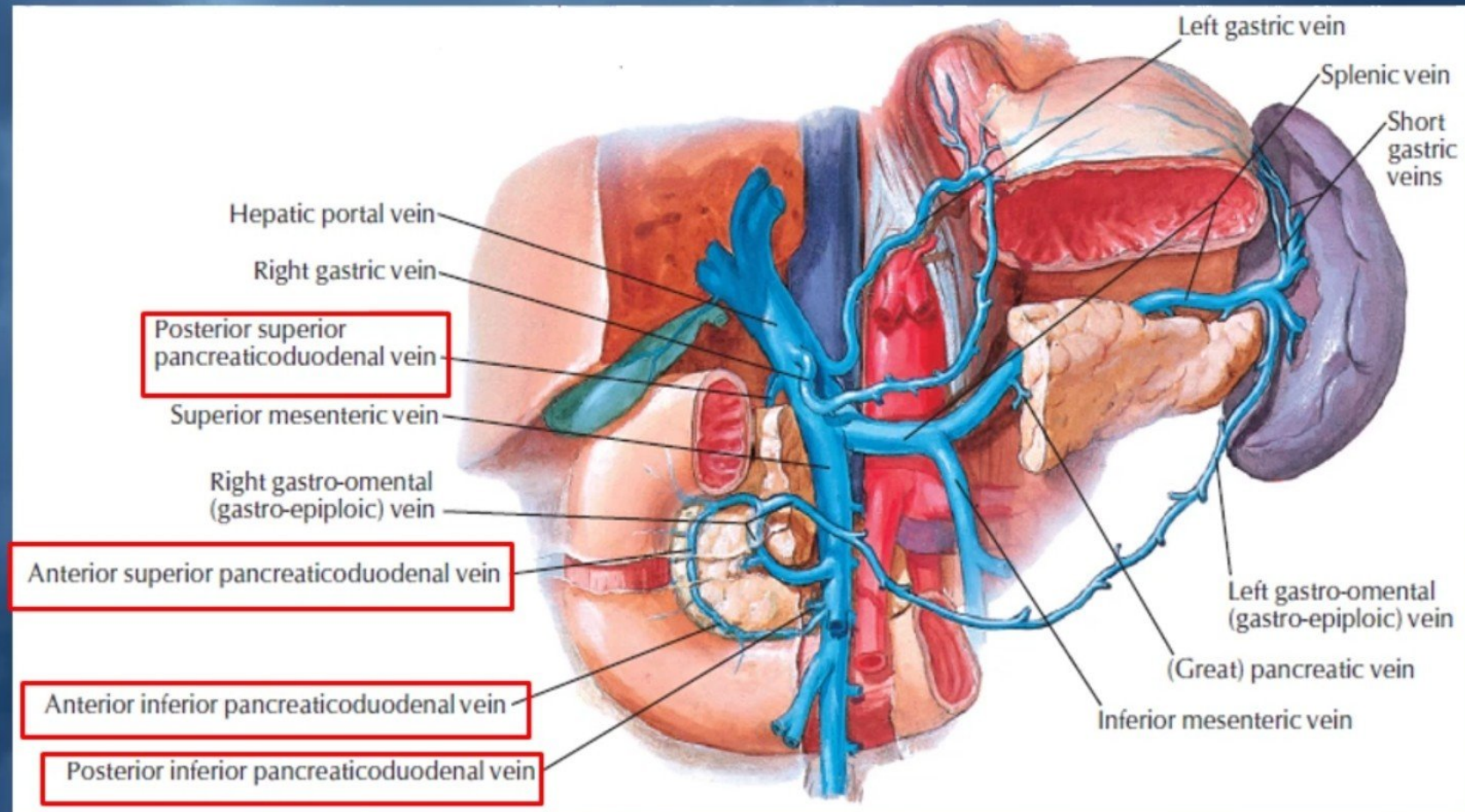
The arterial supply to the duodenum

- Branches from the gastroduodenal artery.
- The supraduodenal artery from the gastroduodenal artery.
- The duodenal branches from the anterior superior pancreaticoduodenal artery (from the gastroduodenal artery).
- The duodenal branches from the posterior superior pancreaticoduodenal artery (from the gastroduodenal artery).
- The duodenal branches from the anterior inferior pancreaticoduodenal artery (from the inferior pancreaticoduodenal artery – a branch of **superior mesenteric artery**).
- The duodenal branches from the posterior inferior pancreaticoduodenal artery (from the inferior pancreaticoduodenal artery – a branch of **superior mesenteric artery**).
- The **first jejunal branch** from the **superior mesenteric artery**.



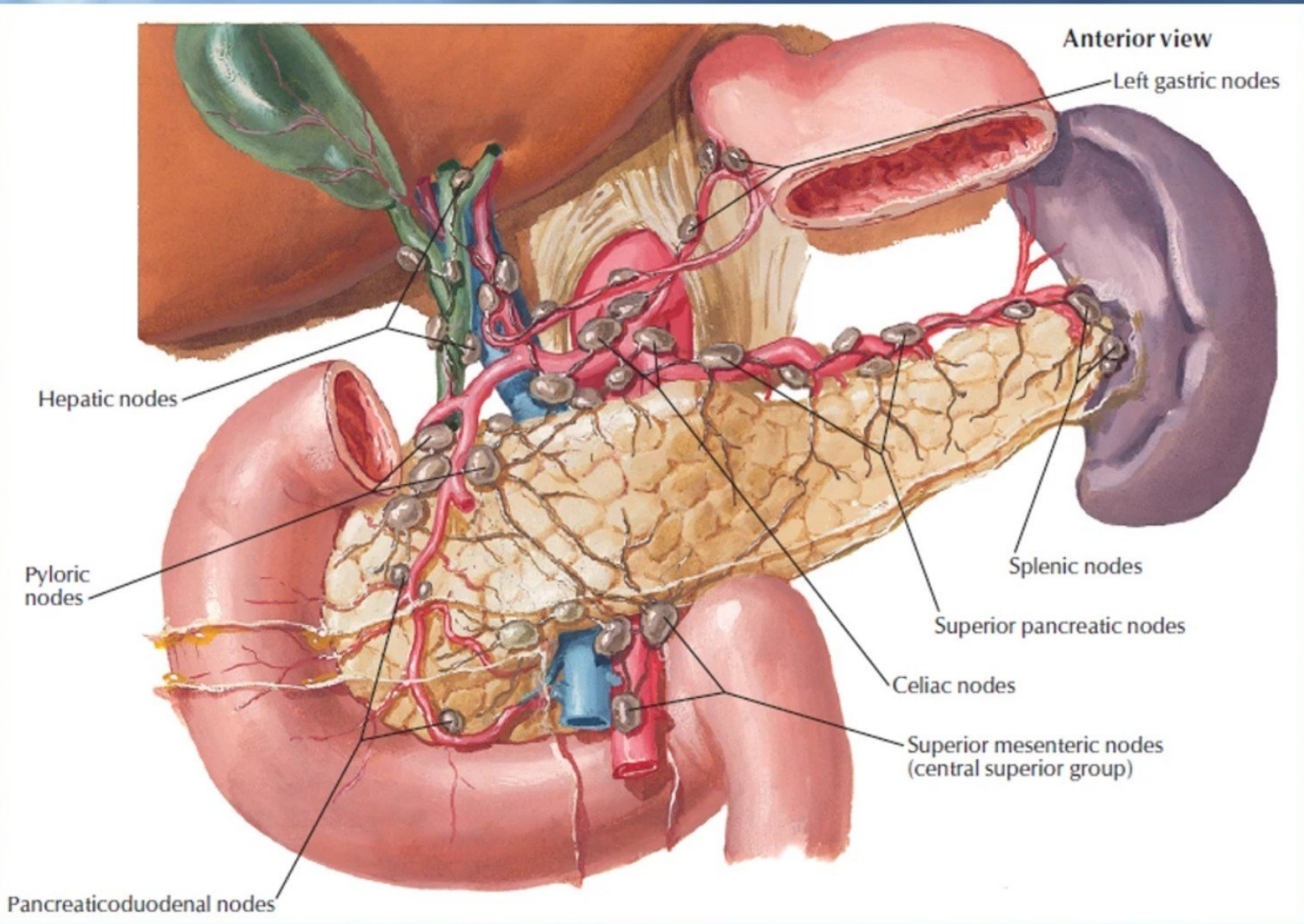
Veins of Duodenum

- The **superior pancreaticoduodenal vein** drains into the **portal vein**.
- The **inferior pancreaticoduodenal vein** joins the **superior mesenteric vein**.



Duodenal Lymph drainage

Duodenal lymph drains by channels that accompany **the superior and inferior pancreaticoduodenal** vessels to **coeliac** and **superior mesenteric nodes**.



Will be continue